

Lecture Notes: The Integers and Number Theory

Discrete Mathematics

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Section 1 Divisibility and Prime Numbers

1.1 Divisibility

Definition (Divisibility). Let $a, b \in \mathbb{Z}$. We say that a *divides* b , denoted $a \mid b$, if there exists an integer c such that $b = ac$. If a does not divide b , we write $a \nmid b$.

Remark. We often say " a is a divisor of b ", " b is a multiple of a ", or " b is divisible by a ". By convention, $a \mid 0$ for every non-zero integer a , but $0 \mid b$ only if $b = 0$.

Theorem (Properties of Divisibility). Let $a, b, c \in \mathbb{Z}$.

- (i) Reflexivity: $a \mid a$ (for $a \neq 0$).
- (ii) Transitivity: If $a \mid b$ and $b \mid c$, then $a \mid c$.
- (iii) Linearity: If $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for any $m, n \in \mathbb{Z}$.
- (iv) If $a \mid b$ and $b \mid a$, then $a = \pm b$.
- (v) If $a \mid b$ and $a, b > 0$, then $a \leq b$.

$3 \mid 12$ because $12 = 3 \cdot 4$. Also, $7 \mid -21$ because $-21 = 7 \cdot (-3)$. However, $5 \nmid 12$ since no integer c satisfies $12 = 5c$.

Since $4 \mid 24$ and $4 \mid 36$, by linearity $4 \mid (2 \cdot 24 + 3 \cdot 36) = 48 + 108 = 156$, and indeed $156 = 4 \cdot 39$.

1.2 Prime and Composite Numbers

Definition (Prime Number). A positive integer $p > 1$ is called *prime* if its only positive divisors are 1 and p itself. A positive integer $n > 1$ that is not prime is called *composite*. The integer 1 is neither prime nor composite; it is called a *unit*.

Examples of primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, ... **Examples of composites:** 4, 6, 8, 9, 10, 12, ...

Theorem (Fundamental Theorem of Arithmetic). Every integer $n > 1$ can be expressed uniquely as a product of prime numbers, up to the order of the factors. That is, there exist unique primes $p_1 < p_2 < \dots < p_k$ and unique positive integers $e_1, e_2, \dots, e_k$ such that

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}.$$

$60 = 2^2 \cdot 3 \cdot 5$. The prime factorization is unique: no other combination of primes multiplied together yields 60.

$84 = 2^2 \cdot 3 \cdot 7$, and $100 = 2^2 \cdot 5^2$.

Theorem (Euclid's Theorem). There are infinitely many prime numbers.

Proof Sketch. Suppose there were only finitely many primes $p_1, p_2, \dots, p_k$. Consider $N = p_1 p_2 \cdots p_k + 1$. Then N is not divisible by any of the listed primes (it leaves remainder 1), so N must be prime itself or divisible by a new prime not in the list, a contradiction.

Section 2 Modular Arithmetic

Definition (Congruence). Let $a, b \in \mathbb{Z}$ and let $n \in \mathbb{Z}^+$ (a positive integer). We say that a is *congruent* to b modulo n , denoted

$$a \equiv b \pmod{n},$$

if $n \mid (a - b)$. The integer n is called the *modulus*.

Remark. $a \equiv b \pmod{n}$ means that a and b have the same remainder upon division by n . Equivalently, $a = b + kn$ for some integer k .

$17 \equiv 5 \pmod{12}$ because $17 - 5 = 12$, which is divisible by 12. Also, $23 \equiv 2 \pmod{7}$ because $23 - 2 = 21 = 3 \cdot 7$.

$-8 \equiv 2 \pmod{5}$ because $-8 - 2 = -10 = -2 \cdot 5$. Note that negative numbers are perfectly valid in modular arithmetic.

2.1 Properties of Congruences

Theorem (Basic Properties). For a fixed modulus n , congruence modulo n is an equivalence relation on $\mathbb{Z}$:

- (i) Reflexivity: $a \equiv a \pmod{n}$.
- (ii) Symmetry: If $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$.
- (iii) Transitivity: If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

Theorem (Arithmetic Operations). Suppose $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then:

$$\begin{aligned} a + c &\equiv b + d \pmod{n}, \\ a - c &\equiv b - d \pmod{n}, \\ ac &\equiv bd \pmod{n}. \end{aligned}$$

Remark. Congruences can be added, subtracted, and multiplied just like equations. However, division is more delicate and is not always valid (it requires coprime conditions).

Compute $7 \cdot 9 + 4 \cdot 11$ modulo 5.

Solution. Reduce first:

$$\begin{aligned} 7 &\equiv 2 \pmod{5}, & 9 &\equiv 4 \pmod{5}, \\ 4 &\equiv 4 \pmod{5}, & 11 &\equiv 1 \pmod{5}. \end{aligned}$$

Then:

$$7 \cdot 9 + 4 \cdot 11 \equiv 2 \cdot 4 + 4 \cdot 1 = 8 + 4 = 12 \equiv 2 \pmod{5}.$$

Section 3 Euclidean Algorithm and GCD

3.1 Greatest Common Divisor

Definition (Greatest Common Divisor). Let $a, b \in \mathbb{Z}$, not both zero. The *greatest common divisor* of a and b , denoted $\gcd(a, b)$, is the largest positive integer d that divides both a and b .

Definition (Coprime). Two integers a and b are said to be *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

$\gcd(24, 36) = 12$, because the common divisors of 24 and 36 are 1, 2, 3, 4, 6, 12, and the largest is 12.

$\gcd(14, 25) = 1$, so 14 and 25 are coprime.

3.2 The Euclidean Algorithm

The Euclidean algorithm is an efficient method for computing $\gcd(a, b)$ based on the following principle:

Theorem (Division Algorithm). For any integers a and b with $b > 0$, there exist unique integers q (the quotient) and r (the remainder) such that

$$a = bq + r, \quad 0 \leq r < b.$$

Lemma. $\gcd(a, b) = \gcd(b, r)$, where r is the remainder when a is divided by b .

Algorithm (Euclidean Algorithm). To compute $\gcd(a, b)$ with $a \geq b > 0$:

Step 1: Set $r_0 = a$, $r_1 = b$.

Step 2: For $i = 1, 2, 3, \dots$, compute r_{i+1} as the remainder when r_{i-1} is divided by r_i :

$$r_{i-1} = q_i r_i + r_{i+1}, \quad 0 \leq r_{i+1} < r_i.$$

Step 3: Stop when $r_{k+1} = 0$. Then $\gcd(a, b) = r_k$, the last non-zero remainder.

Compute $\gcd(252, 105)$ using the Euclidean algorithm.

Solution.

$$252 = 2 \cdot 105 + 42$$

$$105 = 2 \cdot 42 + 21$$

$$42 = 2 \cdot 21 + 0.$$

The last non-zero remainder is 21, so $\gcd(252, 105) = 21$.

3.3 Extended Euclidean Algorithm

The **Extended Euclidean Algorithm** not only computes $\gcd(a, b)$ but also finds integers x and y (called *Bézout coefficients*) such that

$$ax + by = \gcd(a, b).$$

This is done by back-substituting through the steps of the Euclidean algorithm.

Find integers x, y such that $252x + 105y = \gcd(252, 105)$.

Solution. From the previous example, we have:

$$42 = 252 - 2 \cdot 105,$$

$$21 = 105 - 2 \cdot 42.$$

Substitute the first into the second:

$$\begin{aligned} 21 &= 105 - 2 \cdot (252 - 2 \cdot 105) \\ &= 105 - 2 \cdot 252 + 4 \cdot 105 \\ &= (-2) \cdot 252 + 5 \cdot 105. \end{aligned}$$

Thus, $x = -2$, $y = 5$.

Section 4 Solving Congruence Relations

4.1 Linear Congruences

Definition (Linear Congruence). A *linear congruence* is a congruence of the form

$$ax \equiv b \pmod{n},$$

where $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$ are given, and x is an unknown integer.

Theorem (Existence of Solutions). The linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if $\gcd(a, n) \mid b$.

Theorem (Number of Solutions). If $d = \gcd(a, n)$ and $d \mid b$, then the congruence $ax \equiv b \pmod{n}$ has exactly d incongruent solutions modulo n .

4.2 Method of Solving

To solve $ax \equiv b \pmod{n}$ when $d = \gcd(a, n) \mid b$:

Step 1: Divide the entire congruence by d :

$$\frac{a}{d}x \equiv \frac{b}{d} \pmod{\frac{n}{d}}.$$

Now $\gcd(\frac{a}{d}, \frac{n}{d}) = 1$, so $\frac{a}{d}$ has a multiplicative inverse modulo $\frac{n}{d}$.

Step 2: Find the inverse u such that $u \cdot \frac{a}{d} \equiv 1 \pmod{\frac{n}{d}}$ (using the Extended Euclidean Algorithm).

Step 3: Multiply both sides by u to isolate x :

$$x \equiv u \cdot \frac{b}{d} \pmod{\frac{n}{d}}.$$

This gives one solution x_0 modulo $\frac{n}{d}$.

Step 4: The d solutions modulo n are:

$$x \equiv x_0 + k \cdot \frac{n}{d} \pmod{n}, \quad k = 0, 1, 2, \dots, d-1.$$

Solve $14x \equiv 8 \pmod{22}$.

Solution.

- $\gcd(14, 22) = 2$, and $2 \mid 8$, so solutions exist.
- Divide by 2: $7x \equiv 4 \pmod{11}$.
- Find the inverse of 7 modulo 11. Using the extended Euclidean algorithm or trial: $7 \cdot 8 = 56 \equiv 1 \pmod{11}$ (since $56 = 5 \cdot 11 + 1$). So the inverse is 8.
- Multiply: $x \equiv 8 \cdot 4 = 32 \equiv 10 \pmod{11}$.
- The two solutions modulo 22 are:

$$x \equiv 10 \pmod{22}, \quad x \equiv 10 + 11 = 21 \pmod{22}.$$

Check: $14 \cdot 10 = 140 = 6 \cdot 22 + 8 \equiv 8 \pmod{22}$, and $14 \cdot 21 = 294 = 13 \cdot 22 + 8 \equiv 8 \pmod{22}$.

Section 5 Chinese Remainder Theorem

Theorem (Chinese Remainder Theorem, CRT). Let $n_1, n_2, \dots, n_k$ be positive integers that are pairwise coprime (i.e., $\gcd(n_i, n_j) = 1$ for all $i \neq j$). Then the system of congruences

$$\begin{aligned} x &\equiv a_1 \pmod{n_1}, \\ x &\equiv a_2 \pmod{n_2}, \\ &\vdots \\ x &\equiv a_k \pmod{n_k}, \end{aligned}$$

has a unique solution modulo $N = n_1 n_2 \cdots n_k$. That is, there exists a unique x with $0 \leq x < N$ satisfying all congruences simultaneously.

5.1 Constructive Method of Solution

Let $N = n_1 n_2 \cdots n_k$. For each $i = 1, \dots, k$:

Step 1: Compute $N_i = N/n_i$.

Step 2: Find the modular inverse M_i of N_i modulo n_i (i.e., $M_i N_i \equiv 1 \pmod{n_i}$).

Then the solution is:

$$x \equiv a_1 M_1 N_1 + a_2 M_2 N_2 + \cdots + a_k M_k N_k \pmod{N}.$$

Solve the system:

$$x \equiv 2 \pmod{3},$$

$$x \equiv 3 \pmod{5},$$

$$x \equiv 2 \pmod{7}.$$

Solution. Here $n_1 = 3$, $n_2 = 5$, $n_3 = 7$, which are pairwise coprime. $N = 3 \cdot 5 \cdot 7 = 105$.

– $N_1 = 105/3 = 35$. Inverse of 35 mod 3: $35 \equiv 2 \pmod{3}$, and $2 \cdot 2 = 4 \equiv 1 \pmod{3}$, so $M_1 = 2$.

– $N_2 = 105/5 = 21$. Inverse of 21 mod 5: $21 \equiv 1 \pmod{5}$, so $M_2 = 1$.

– $N_3 = 105/7 = 15$. Inverse of 15 mod 7: $15 \equiv 1 \pmod{7}$, so $M_3 = 1$.

Then:

$$\begin{aligned} x &\equiv 2 \cdot 2 \cdot 35 + 3 \cdot 1 \cdot 21 + 2 \cdot 1 \cdot 15 \\ &= 140 + 63 + 30 = 233 \pmod{105}. \end{aligned}$$

Since $233 = 2 \cdot 105 + 23$, we have $x \equiv 23 \pmod{105}$. Check: $23 \equiv 2 \pmod{3}$, $23 \equiv 3 \pmod{5}$, $23 \equiv 2 \pmod{7}$.

Solve:

$$x \equiv 1 \pmod{4},$$

$$x \equiv 2 \pmod{9}.$$

$N = 36$. $N_1 = 9$, inverse of 9 mod 4: $9 \equiv 1$, so $M_1 = 1$. $N_2 = 4$, inverse of 4 mod 9: $4 \cdot 7 = 28 \equiv 1$, so $M_2 = 7$. Then:

$$x \equiv 1 \cdot 1 \cdot 9 + 2 \cdot 7 \cdot 4 = 9 + 56 = 65 \equiv 29 \pmod{36}.$$

Section 6 Diophantine Equations

Definition (Diophantine Equation). A *Diophantine equation* is a polynomial equation in which only integer solutions are sought. In this section, we focus on *linear Diophantine equations* in two variables.

Definition (Linear Diophantine Equation). An equation of the form

$$ax + by = c,$$

where $a, b, c \in \mathbb{Z}$ are given, and $x, y \in \mathbb{Z}$ are unknowns.

6.1 Conditions for Solutions

Theorem (Existence of Solutions). The linear Diophantine equation $ax + by = c$ has integer solutions if and only if $\gcd(a, b) \mid c$.

Proof Sketch. Let $d = \gcd(a, b)$. For any integers x, y , the left-hand side $ax + by$ is always a multiple of d . So c must be a multiple of d for solutions to exist. Conversely, if $d \mid c$, the extended Euclidean algorithm gives a particular solution.

6.2 Finding the General Solution

Suppose $d = \gcd(a, b) \mid c$. The method:

Step 1: Use the Extended Euclidean Algorithm to find one particular solution (x_0, y_0) to $ax + by = d$.

Step 2: Multiply by c/d to obtain a particular solution to $ax + by = c$:

$$x_p = \frac{c}{d}x_0, \quad y_p = \frac{c}{d}y_0.$$

Step 3: The general solution is given by:

$$\begin{cases} x = x_p + \frac{b}{d} \cdot t, \\ y = y_p - \frac{a}{d} \cdot t, \end{cases} \quad t \in \mathbb{Z}.$$

Find all integer solutions to $14x + 22y = 50$.

Solution. First, $\gcd(14, 22) = 2$, and $2 \mid 50$, so solutions exist.

– Find one solution to $14x + 22y = 2$ using the Extended Euclidean Algorithm.

$$22 = 1 \cdot 14 + 8,$$

$$14 = 1 \cdot 8 + 6,$$

$$8 = 1 \cdot 6 + 2,$$

$$6 = 3 \cdot 2 + 0.$$

Back-substituting:

$$2 = 8 - 1 \cdot 6,$$

$$6 = 14 - 1 \cdot 8 \implies 2 = 8 - 1 \cdot (14 - 1 \cdot 8) = 2 \cdot 8 - 1 \cdot 14,$$

$$8 = 22 - 1 \cdot 14 \implies 2 = 2 \cdot (22 - 1 \cdot 14) - 1 \cdot 14 = 2 \cdot 22 - 3 \cdot 14.$$

So $(-3) \cdot 14 + 2 \cdot 22 = 2$. Thus $x_0 = -3, y_0 = 2$.

– Multiply by $c/d = 50/2 = 25$: $x_p = 25 \cdot (-3) = -75, y_p = 25 \cdot 2 = 50$.

– General solution:

$$x = -75 + \frac{22}{2}t = -75 + 11t, \quad y = 50 - \frac{14}{2}t = 50 - 7t, \quad t \in \mathbb{Z}.$$

Check: $14(-75 + 11t) + 22(50 - 7t) = -1050 + 154t + 1100 - 154t = 50$.

Find all integer solutions to $6x + 9y = 21$.

Solution. $\gcd(6, 9) = 3$, and $3 \mid 21$, so solutions exist. Divide the equation by 3 to simplify: $2x + 3y = 7$. By inspection, a particular solution to $2x + 3y = 1$ is $(x_0, y_0) = (-1, 1)$, since $2(-1) + 3(1) = 1$. Multiply by 7: $x_p = -7, y_p = 7$. General solution:

$$x = -7 + \frac{3}{1}t = -7 + 3t, \quad y = 7 - \frac{2}{1}t = 7 - 2t, \quad t \in \mathbb{Z}.$$

Remark. Diophantine equations are named after the ancient Greek mathematician Diophantus of Alexandria. Linear Diophantine equations in two variables are completely solved by the method above. Higher-degree Diophantine equations (e.g., $x^2 + y^2 = z^2$, $x^n + y^n = z^n$) are much more challenging and have led to deep mathematical theories, including Fermat's Last Theorem.